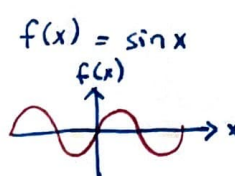
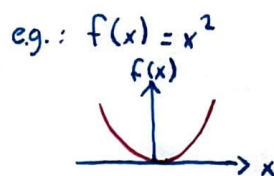
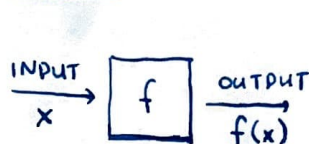
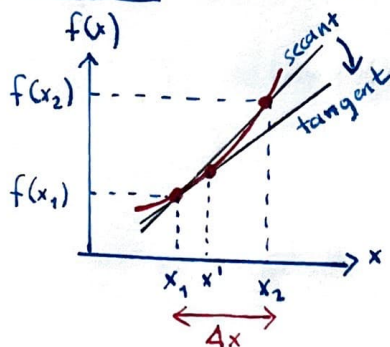


Functions



! An input cannot be assigned to two different outputs at the same time:
 $f(x)$ is not a function

Derivative: How much $f(x)$ changes when we change x a little bit?



$$\lim_{\Delta x \rightarrow 0} \frac{f(x_2) - f(x_1)}{\Delta x} \equiv \left. \frac{df}{dx} \right|_{x=x_1} \equiv f'$$

: Slope of the tangent line at x_1

As x_2 gets closer to x_1 , secant line gets closer to tangent line at x_1

Derivatives of common functions:

- Power functions: $f(x) = x^n$, n constant $\frac{df}{dx} = n x^{n-1}$

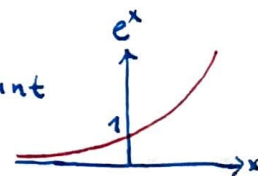
e.g. $f(x) = c$, c constant $\Rightarrow f' = 0$

$f(x) = x \Rightarrow f' = 1$

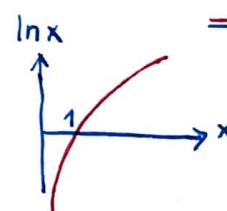
$f(x) = x^2 \Rightarrow f' = 2x$

$f(x) = \sqrt{x} \Rightarrow f' = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}$

- Exponential function: $f(x) = e^x$, $e = 2.71...$ mathematical constant $f' = e^x$



- Logarithms: $f(x) = \ln x \equiv \log_e x \Rightarrow f' = 1/x$
 $g(x) = \log_a x \Rightarrow g' = 1/x \ln a$



- Trigonometry:
 $f(x) = \sin x \quad f' = \cos x$
 $g(x) = \cos x \quad g' = -\sin x$
 $h(x) = \tan x \quad h' = \sec^2 x$
 $k(x) = \cot x \quad k' = -\csc^2 x$
 $\sec x = \frac{1}{\cos x}$
 $\csc x = \frac{1}{\sin x}$

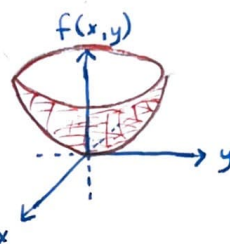
- Inverse Trigonometry:
 $f(x) = \arcsin x \quad f' = 1/\sqrt{1-x^2}$
 $g(x) = \arccos x \quad g' = -1/\sqrt{1-x^2}$
 $h(x) = \arctan x \quad h' = 1/(1+x^2)$
 $k(x) = \text{arccot } x \quad k' = -1/(1+x^2)$

Derivative Rules:

- $\frac{d}{dx} [cf(x)] = cf'(x)$: Multiplication by constant c
 - $\frac{d}{dx} [f(x) \pm g(x)] = f'(x) \pm g'(x)$: Sum rule
 - $\frac{d}{dx} [f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$: ~~Sum~~ Product rule
 - $\frac{d}{dx} \{f[g(x)]\} = f'[g(x)]g'(x)$: Chain rule
- e.g. $(2x+1)^2 \rightarrow \frac{d}{dx} [(2x+1)^2] = 2(2x+1) \cdot 2$
- Corollary: $\frac{d}{dx} \left[\frac{f}{g} \right] = \frac{d}{dx} [fg^{-1}] = f'g^{-1} + (-1)g^{-2}fg' = \frac{f'g - fg'}{g^2}$
 $\frac{d}{dx} \left[\frac{1}{f} \right] = \frac{d}{dx} [f^{-1}] = -f^{-2}f' = -\frac{f'}{f^2}$

Partial derivatives of multivariable functions:

$f = f(x, y, z, \dots)$ e.g. $f(x, y) = x^2 + y^2 \rightarrow$



A surface in 3D space.

- Partial derivative: Derivative of a multi. var. function with respect to one of its parameters.

e.g. $f(x, y) = x^2 + y^2 \Rightarrow \frac{\partial f}{\partial x} = 2x$, $g(x, y) = x^2 + xy^2 \Rightarrow \frac{\partial g}{\partial x} = 2x + y^2$

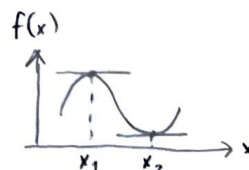
- Total differential: The smallest change in the value of a multivariable function when we change all of its parameters a little bit.

e.g. $f = f(x, y, z) \Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz$

- Total derivative: $f = f(x_1, x_2, \dots, x_n, t)$ and $x_i(t) = x_i$ where $i = 1, \dots, n$

$$\frac{df}{dt} = \sum_{i=1}^n \left(\frac{\partial f}{\partial x_i} \frac{dx_i}{dt} \right) + \frac{\partial f}{\partial t}$$

- Maximum and minimum value of a function:

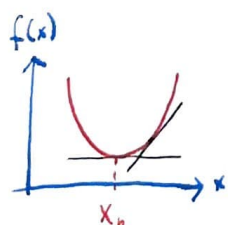


slopes of tangential lines at x_1 and x_2 are zero.

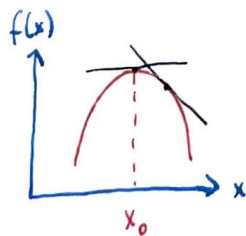
If we have a function $f(x)$ and we want to find the x value where the function takes its minimum or maximum value;

$\frac{d}{dx} [f(x)] = 0$: we solve this equation for x .

- Concave upward and downward: If a function has a maximum/minimum at some point x_0 , how can we understand that the function is concave up or down at the same x_0 ?



slope is increasing



slope is decreasing.

$\frac{d^2 f(x)}{dx^2} > 0$: concave up

$\frac{d^2 f(x)}{dx^2} < 0$: concave down

Taylor series: If we are interested how a given function $f(x)$ behaves in the neighborhood of a particular point x_0 , we use Taylor series.

- Theorem: An analytic function can ^{locally} be written as a convergent power series.

Analytic functions: all polynomials, exponential function, trigonometric functions, logarithm, power functions

How to understand if a given function is analytic? : It must be infinitely differentiable.

$$f(x) = a_0 + a_1(x-x_0) + a_2(x-x_0)^2 + a_3(x-x_0)^3 + a_4(x-x_0)^4 + \dots$$

The series is convergent to $f(x)$ for x in the neighborhood of x_0 .

How can we determine the coefficients a_i ?

$$f'(x) = a_1 + 2a_2(x-x_0) + 3a_3(x-x_0)^2 + 4a_4(x-x_0)^3 + \dots$$

$$f''(x) = 2a_2 + 3 \cdot 2a_3(x-x_0) + 4 \cdot 3a_4(x-x_0)^2 + \dots$$

$$f'''(x) = 3 \cdot 2a_3 + 4 \cdot 3 \cdot 2a_4(x-x_0) + \dots$$

$$f^{(4)}(x) = 4 \cdot 3 \cdot 2a_4 + \dots$$

We know that $f(x_0) = a_0$, $f'(x_0) = a_1$, $f''(x_0) = 2a_2$, $f'''(x_0) = 3 \cdot 2 \cdot a_3$, $f^{(4)}(x_0) = 4 \cdot 3 \cdot 2 \cdot a_4$

$$\Rightarrow a_1 = f'(x_0), a_2 = f''(x_0)/2, a_3 = f'''(x_0)/3!, a_4 = f^{(4)}(x_0)/4!$$

$$\Rightarrow f(x) = a_0 + f'(x_0)(x-x_0) + \frac{f''(x_0)}{2!}(x-x_0)^2 + \frac{f'''(x_0)}{3!}(x-x_0)^3 + \frac{f^{(4)}(x_0)}{4!}(x-x_0)^4 + \dots$$

THIS IS CALLED TAYLOR SERIES!

Integral: Calculation of anti-derivatives

• $f(x) = x^2 \Rightarrow f' = 2x \equiv g(x) \rightarrow \int g(x) dx = x^2 + C$ family of functions

• $f(x) = \cos x \Rightarrow \int f(x) dx = \sin x + C$

• $f(x) = 1/x \Rightarrow \int f(x) dx = \ln x + C$

Indefinite integrals: results in a function.

Definite integral: $\int_a^b f(x) dx = F(b) - F(a)$ where $F(x)$ is the antiderivative of $f(x)$.

e.g. $\int_1^2 x^2 dx = \left. \frac{x^3}{3} \right|_1^2 = \frac{8}{3} - \frac{1}{3} = \frac{7}{3}$: Definite integral results in a number.

Properties of the definite integral:

INDEF ✓ • $\int_a^b f(x) dx = \int_a^b f(t) dt$

• $\int_a^b f(x) dx = - \int_b^a f(x) dx$

INDEF ✓ • $\int_a^b c f(x) dx = c \int_a^b f(x) dx$ constant

INDEF ✓ • $\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$

• $\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$

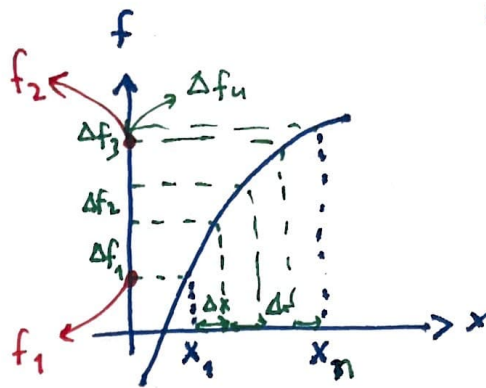
• $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$ if $f(x)$ is an even function

• $\int_{-a}^a f(x) dx = 0$ if $f(x)$ is an odd function.

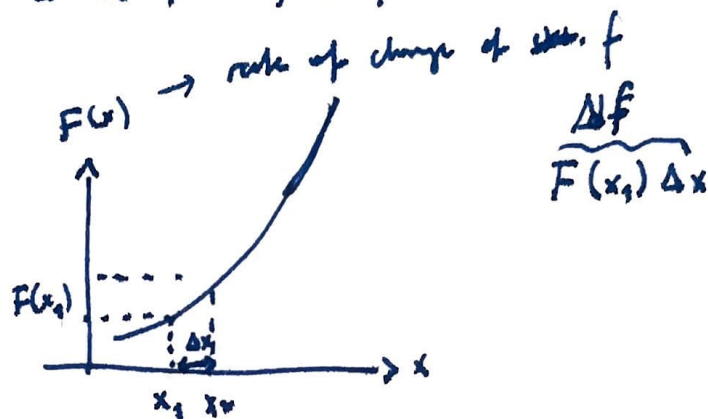
Integral

$$\frac{\Delta f}{\Delta x} \Delta x = \Delta f, f_2 - f_1$$

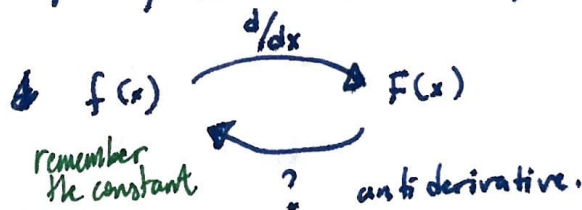
$$\sum_{i=1}^n \Delta f_i \approx f_2 - f_1$$



If we have the function $\hat{F}(x)$ indicating the rate of change of another function $f(x)$, how do we find $f(x)$?



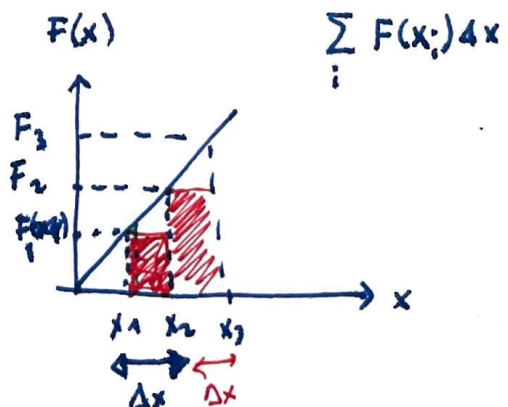
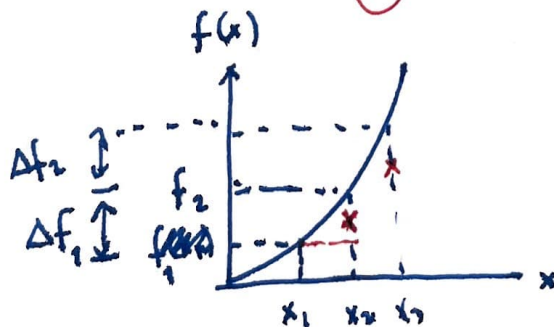
Assume that we are given a function $F(x)$ and we know that it reflects the rate of change of another function $f(x)$ with respect to x .



In other words

$\hat{f}(x)$ we know how much $f(x)$ changes per unit x . We are wondering how much $f(x)$ changes

$$\int_{x_1}^{x_2} \frac{\Delta f}{\Delta x} \Delta x \approx \Delta f = f_2 - f_1$$

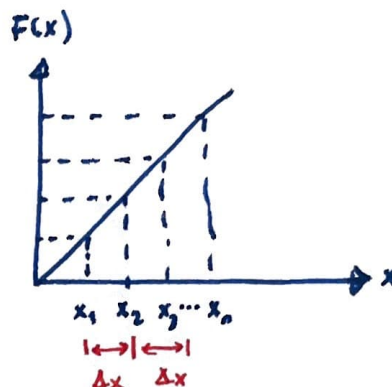
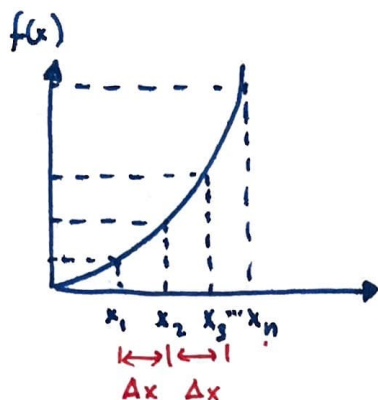


* The integral of a derivative over an interval is given by the value of the function at end points.

Integral

Assume that we are given a function $F(x)$ and we know that it reflects the rate of change of another function $f(x)$ with respect to x .

$$\frac{df}{dx} dx = df$$



$$\frac{\Delta f}{\Delta x} \Delta x \approx \Delta f$$

$$F(x_1) \Delta x + F(x_2) \Delta x$$

$$f(x_n) - f(x_1) \approx \sum_{i=1}^n F(x_i) \Delta x$$

$$f(x_n) - f(x_1)$$

$$f(x_2) - f(x_1) = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n F(x_i) \Delta x$$

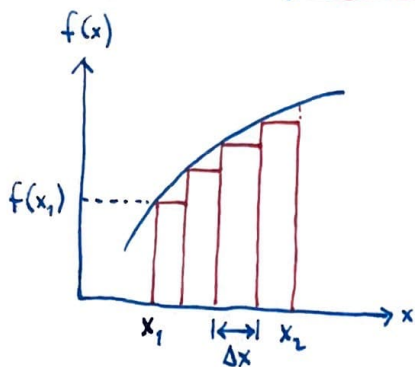
$$= \int_{x_1}^{x_2} F(x) dx \quad \text{definite integral}$$

• Once we are given $F(x)$ we know how to find $f(x) \rightarrow$ anti-derivative

• Notation: ~~An~~ Indefinite integrals

$$\int F(x) dx : \text{anti-derivative.}$$

Definition and geometric interpretation of definite integral:



What is the area under $f(x)$ between x_1 and x_2 ?

$$\Delta A_1 = f(x_1) \Delta x$$

$$\Delta A_i = f(x_i) \Delta x$$

$\sum_i f(x_i) \Delta x \approx$ Approximate area under $f(x)$ between x_1 and x_2 .

$$\lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(x_i) \Delta x = \int_{x_1}^{x_2} f(x) dx \quad \text{: Definition of integral.}$$

linear

First order differential equations:

An equation in the form $F(x, y, y') = 0$

The solution $f(x)$ satisfies $F[x, f(x), f'(x)] = 0$ for every value of x .

Example: $y' = x^2 + 1 \rightarrow y' - x^2 - 1 = 0 \rightarrow$ in the form $F(x, y, y') = 0$

$$\frac{dy}{dx} = x^2 + 1 \rightarrow dy = x^2 dx + dx \rightarrow \int dy = \int (x^2 dx + dx) + C$$

$$\Rightarrow y(x) = \frac{x^3}{3} + x + C$$

Second order linear differential equations:

General form: $y'' + p(x)y' + q(x)y = f(x)$

We will not consider the general solution of this equation for now. We are going to just consider some particular cases which we will need in the future.

• Case I: $p(x) = q(x) = 0$ and $f(x) = C$, where C is a ^{fixed} constant that we know.

$$y'' = C \rightarrow \frac{d}{dx}[y'] = C \rightarrow dy' = C dx \rightarrow \int dy' = C \int dx + D \rightarrow \text{an arbitrary constant}$$

$$\rightarrow y'(x) = Cx + D \rightarrow \frac{dy}{dx} = Cx + D \rightarrow dy = Cx dx + D dx \rightarrow \int dy = C \int x dx + D \int dx + E$$

$$\rightarrow y(x) = \frac{Cx^2}{2} + Dx + E$$

We have two random constants, D and E .

To completely solve the problem, we need "initial conditions"

$y(0) = A$
 $y'(0) = B$ } Initial conditions $\rightarrow$ we assume here that A and B are some fixed constants that we know.

$$\Rightarrow y(0) = E = A$$

$$y'(0) = D = B$$

$\Rightarrow$ The exact solution with the given initial conditions is

$$y(x) = \frac{Cx^2}{2} + Bx + A$$

- Case II: $p(x) = C$, $q(x) = 0$ and $f(x) = D$, C and D are constants.

$$y'' + Cy' = D \rightarrow y'' = D - Cy' \rightarrow \frac{dy'}{dx} = D - Cy' \rightarrow \frac{dy'}{D - Cy'} = dx$$

$$\boxed{\begin{matrix} y(0) = y_0 \\ y'(0) = y'_0 \end{matrix}} \text{ Initial conditions}$$

- Change variables:

$$D - Cy' = h \rightarrow \frac{dh}{dy'} = -C \rightarrow dy' = -\frac{dh}{C}$$

- Substitute:

$$-\frac{1}{C} \frac{dh}{h} = dx \rightarrow \int dx = -\frac{1}{C} \int \frac{dh}{h} + \text{arbitrary constant } M$$

$$\rightarrow x - M = -\frac{1}{C} \ln h \xrightarrow{\text{switch back to original variable}} x - M = -\frac{1}{C} \ln[D - Cy']$$

- At this point we can use the initial condition $y'(0) = y'_0$ to find M .

$$0 - M = -\frac{1}{C} \ln[D - Cy'_0] \rightarrow M = \frac{1}{C} \ln(D - Cy'_0)$$

- Exponentiate both sides of $-C[x - M] = \ln[D - Cy']$

$$e^{-C[x-M]} = e^{\ln[D - Cy']} \Rightarrow D - Cy' = e^{-C[x-M]}$$

$$Cy' = D - e^{-C[x-M]} \rightarrow \boxed{y'(x) = \frac{D - e^{-C(x-M)}}{C}}$$

- To find $y(x)$, we repeat integration one more time:

$$\frac{dy}{dx} = \frac{D - e^{-C(x-M)}}{C} \rightarrow \int dy = \frac{1}{C} \int [D - e^{-C(x-M)}] dx + \text{arbitrary constant } N$$

$$\begin{aligned} \Rightarrow y(x) &= \frac{1}{C} \left\{ Dx - \left[-\frac{e^{-C(x-M)}}{C} \right] \right\} + N = \frac{1}{C} \left[Dx + \frac{e^{-C(x-M)}}{C} \right] + N \\ &= \frac{D}{C} x + \frac{e^{-C(x-M)}}{C^2} + N \end{aligned}$$

- Plug in the initial conditions:

$$y(0) = \frac{e^{CM}}{C^2} + N = y_0 \Rightarrow N = y_0 - \frac{e^{CM}}{C^2}$$

$$y(x) = y_0 + \frac{D}{C} x + \frac{e^{-Cx} e^{CM} - e^{CM}}{C^2}$$

$$\boxed{y(x) = y_0 + \frac{D}{C} x + \frac{e^{CM} [e^{-Cx} - 1]}{C^2}}$$

- Case III: $p(x) = 0$, $q(x) = A^2$ and $f(x) = 0$

↪ fixed constant

$y'' + A^2 y = 0$: We know that exponential function solves this diff. equation.

- Solution suggestion: $y(x) = e^{\lambda x}$, $y'(x) = \lambda e^{\lambda x}$, $y''(x) = \lambda^2 e^{\lambda x}$: plug these solutions into the diff. equation.

$$\lambda^2 \cancel{e^{\lambda x}} + A^2 \cancel{e^{\lambda x}} = 0 \rightarrow \lambda^2 = -A^2 \Rightarrow \lambda_{1,2} = \pm iA : \text{roots}$$

- Solutions are $y_1(x) = e^{iAx}$ $y_2(x) = e^{-iAx}$

- The general solution is $y(x) = Cy_1(x) + Dy_2(x) = Ce^{iAx} + De^{-iAx}$

Examples:

- 1) We are given a function $f(x) = 2x^3 + 3\sin[x^2 + x \ln x]$. What is the slope of the tangent line to the curve at $x=1$?

$$\begin{aligned} f'(x) &= 6x^2 + 3\cos[x^2 + x \ln x] \left[2x + \ln x + \frac{1}{x} x \right] \\ &= 6x^2 + 3\cos[x^2 + x \ln x] [2x + \ln x + 1] \end{aligned}$$

$$f'(1) = 6 + 3\cos[1] [2 + 0 + 1] = 6 + 9\cos(1)$$

- 2) For the function $f(x) = 3x^2 + 6x + 1$ estimate the value of the function at $x = 1.01$.

$$f'(x) = 6x + 6 = 6(x+1) \Rightarrow \frac{\Delta f}{\Delta x} = 6(x+1) \rightarrow \Delta f = 6(x+1)\Delta x$$

rate of change of f w.r.t x $\Delta x = 0.01$

$$f(1) = 3 + 6 + 1 = 10$$

$$f(1 + \Delta x) \approx f(1) + f'(1)\Delta x = 10 + 6(1+1) \times 0.01 = 10 + 12 \times 0.01 = 10.12$$

$$\text{Real value: } f(1.01) = 10.1203 \approx 10.12$$

- 3) $f(x, y) = 3x^4 \cos(x^2) + x\sqrt{y}$ what is $\partial f / \partial x$ and $\partial f / \partial y$?

$$\frac{\partial f}{\partial x} = 3[4x^3 \cos(x^2) - \sin(x^2) 2x x^4] + \sqrt{y} = 3[4x^3 \cos(x^2) - 2x^5 \sin(x^2)] + \sqrt{y}$$

$$\frac{\partial f}{\partial y} = 0 + x \left(\frac{1}{2}\right) y^{-1/2} = \frac{x}{2} \frac{1}{\sqrt{y}}$$

- 4) $f(x, y, z) = x^3 + \ln y + \tan(z)$ and $x = t^2 + 1$, $y = 3t + t^3$, $z = \sin(t)$
what is the derivative of $f(x, y, z)$ w.r.t t ? (Total derivative)

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} = \underbrace{3x^2}_{t^2+1} \underbrace{2t}_{\frac{y}{3t+t^3}} + \underbrace{\frac{1}{y}}_{\frac{1}{3t+t^3}} (3+3t^2) + \underbrace{\sec^2 z}_{\sin t} \cos t$$

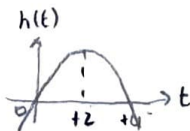
- 5) A particle's height is given by the function $h(t) = [-(t-2)^2 + 4]$ m.
 t is in seconds. If particle starts its motion at $t=0$, after how many seconds will the particle reach its max. height?

$$h(t) = t^2 : \text{U-shape}$$

$$h(t) = -t^2 : \text{inverted U-shape}$$

$$h(t) = -(t-2)^2 : \text{inverted U-shape shifted right}$$

$$h(t) = -(t-2)^2 + 4$$



$$\frac{dh}{dt} = -2(t-2) = 0 \Rightarrow \boxed{t=2}$$

- 6) $V(x) = m(x^3 - 3x)$, $m > 0$ constant. Find the local maximum and minimum of $V(x)$.

$$\frac{dV}{dx} = m(3x^2 - 3) = 0 \Rightarrow 3x^2 = 3 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$$

$$\frac{d^2V}{dx^2} = m6x \Rightarrow \left. \frac{d^2V}{dx^2} \right|_{x=1} = 6m > 0 : \text{concave up} \Rightarrow \text{local minimum}$$

$$\left. \frac{d^2V}{dx^2} \right|_{x=-1} = -6m < 0 : \text{concave down} \Rightarrow \text{local maximum}$$

7) Find Taylor series of e^x , $\sin x$, $\cos x$ around $x=0$. How do these functions behave if $x \ll 1$?

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

when $x \ll 1$:

$$e^x \approx 1+x$$

$$\sin x \approx x$$

$$\cos x \approx 1$$

$$e^{i\theta} = 1 + i\theta + \frac{(i\theta)^2}{2!} + \frac{(i\theta)^3}{3!} + \frac{(i\theta)^4}{4!} + \frac{(i\theta)^5}{5!} + \dots$$

$$= 1 + i\theta - \frac{\theta^2}{2!} - \frac{i\theta^3}{3!} + \frac{\theta^4}{4!} + \frac{i\theta^5}{5!} + \dots$$

$$= \left(1 - \frac{\theta^2}{2!} + \frac{\theta^4}{4!} - \dots\right) + i \left(\theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \dots\right) = \cos \theta + i \sin \theta \rightarrow \text{Euler's formula}$$

MECHANICS

Lecture 1.1: (vectors and operations on them)

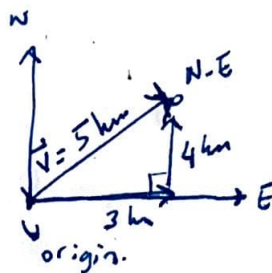
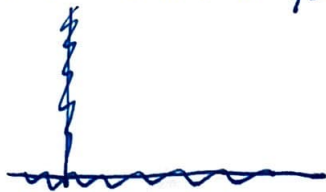
(Definition of position, speed, velocity and acceleration)

Vectors and the operations on them

• Scalar: It's ~~referring to~~ a quantity, e.g., 2 people, 5 kgs of apple, ...

• Vector: It's a quantity associated with a direction in the space.

e.g., drive from London to ~~west~~ north east



$$\vec{V} = \begin{pmatrix} 3 \\ 4 \end{pmatrix} \text{ components}$$

$$|\vec{V}| = \sqrt{3^2 + 4^2} = 5 \text{ km}$$

→ magnitude of a vector

Displacement vector.
Position vector

Unit vector: A vector with a [unit magnitude] = [$|\vec{v}| = 1$]

e.g. $\vec{V} = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$ is a unit vector? $|\vec{V}| = \frac{1}{2} + \frac{1}{2} = 1 \Rightarrow \text{YES!}$

Normalization of a vector: How to obtain a unit vector in an arbitrary direction?

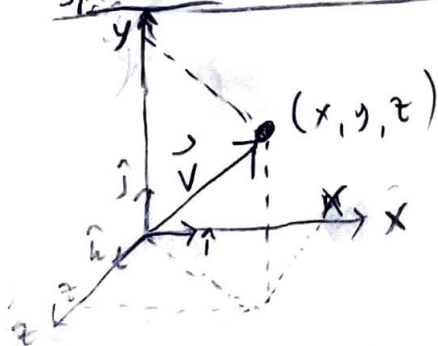
$$\vec{V} = \begin{pmatrix} a \\ b \end{pmatrix} \quad |\vec{V}| = \sqrt{a^2 + b^2}$$

$$\vec{V}' = \begin{pmatrix} a/\sqrt{a^2+b^2} \\ b/\sqrt{a^2+b^2} \end{pmatrix} \quad |\vec{V}'| = \frac{a^2}{a^2+b^2} + \frac{b^2}{a^2+b^2} = \frac{a^2+b^2}{a^2+b^2} = 1$$

$$\vec{V}' = \begin{pmatrix} a/|\vec{V}| \\ b/|\vec{V}| \end{pmatrix}$$

• We will use these notations interchangeably

Special unit vectors in 3 dimension: $(\hat{i}, \hat{j}, \hat{k})$ or $(\hat{e}_1, \hat{e}_2, \hat{e}_3)$



$$\vec{V} = x\hat{i} + y\hat{j} + z\hat{k} : \text{unit vector representation}$$

$$|\vec{V}| = \sqrt{x^2 + y^2 + z^2}$$

$$\hat{i} = (1, 0, 0) \quad \hat{j} = (0, 1, 0) \quad \hat{k} = (0, 0, 1)$$

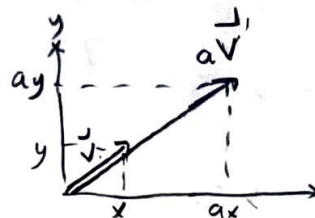
Operations on vectors:

Scalar product:

$$\vec{V} = (x, y, z) = x\hat{i} + y\hat{j} + z\hat{k}$$

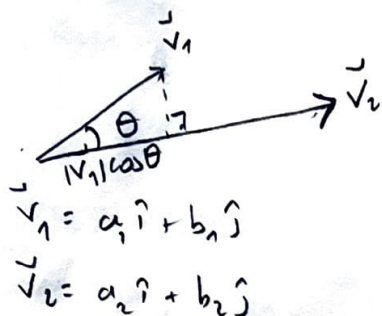
Scalar

$$a\vec{V} = (ax, ay, az) = ax\hat{i} + ay\hat{j} + az\hat{k}$$



$$|a\vec{V}| = \sqrt{a^2(x^2 + y^2 + z^2)} = a|\vec{V}|$$

Dot product:



Geometrical definition:

$$\vec{V}_1 \cdot \vec{V}_2 = |\vec{V}_2| [|\vec{V}_1| \cos \theta] = \text{SCALAR!!!}$$

$$(i) \theta = 0 \Rightarrow |\vec{V}_2| |\vec{V}_1|$$

$$(ii) \theta = \pi/2 \Rightarrow 0$$

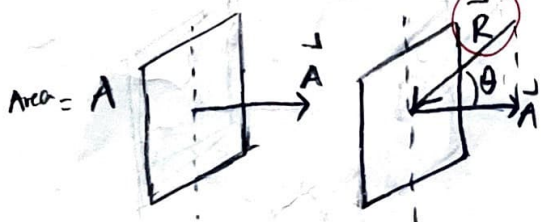
COMMUTATIVE!!!

$$\vec{V}_1 \cdot \vec{V}_2 = a_1 a_2 \underbrace{\hat{i} \cdot \hat{i}}_{\theta=0, 1 \cdot 1=1} + a_1 b_2 \underbrace{\hat{i} \cdot \hat{j}}_{\theta=\pi/2, 0} + b_1 a_2 \underbrace{\hat{j} \cdot \hat{i}}_{\theta=\pi/2, 0} + b_1 b_2 \underbrace{\hat{j} \cdot \hat{j}}_{\theta=0, 1 \cdot 1=1}$$

Example: Rain coming into the living room through the window

Introduce the concept "flux" (back of the page)

$$= a_1 a_2 + b_1 b_2$$



$\theta = \pi/2$ no rain goes inside

$\theta = 0$ whatever rains goes into the window depending on the area of the window frame.

$$|\vec{A}| |\vec{R}| \cos \theta = \vec{R} \cdot \vec{A}$$

$$\theta = \alpha \neq 0 \neq \pi/2$$

Vector product:

$$\vec{V}_1 = a_1 \hat{i} + b_1 \hat{j}$$

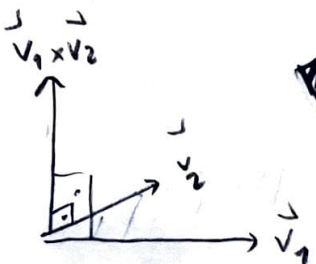
$$\vec{V}_2 = a_2 \hat{i} + b_2 \hat{j}$$

$$|\vec{V}_1 \times \vec{V}_2| = |\vec{V}_1| |\vec{V}_2| \sin \theta \quad \text{magnitude}$$

Direction of $\vec{V}_1 \times \vec{V}_2$ is calculated using the RHT rule.

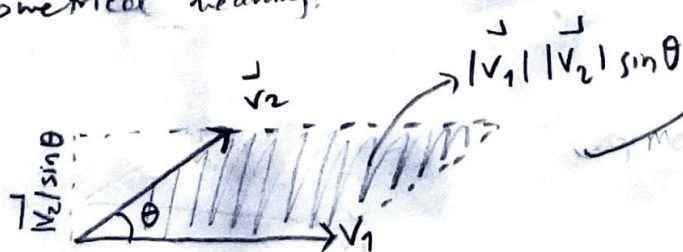
$$\vec{V}_1 \times \vec{V}_2 = \text{result is a vector}$$

$$\vec{V}_1 \times \vec{V}_2 \neq \vec{V}_2 \times \vec{V}_1$$

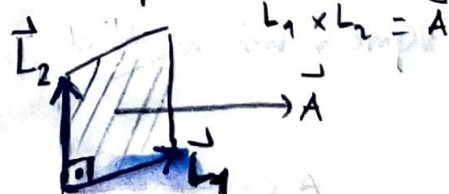


The resulting vector is perpendicular to the plane defined by $\vec{V}_1$ and $\vec{V}_2$.

Geometrical meaning:



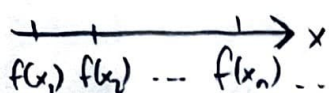
Merge this with rain/window example:



Scalar Field, Vector Field: (It can be skipped)

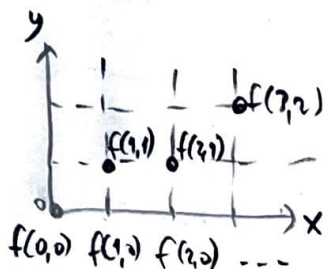
Functions in maths: $f(x)$, $x(t)$, ...
scalar field $\rightarrow$ will be useful when we define the potentials.

[1D]



Associates every number x with another number $f(x) \rightarrow$ scalar field.

[2D]



e.g. temperature of a metal plate at a given time

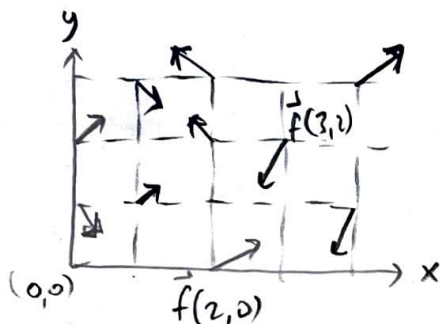
vector field:

[1D]



Associates every number x with a vector $f(x) \rightarrow$ vector field.

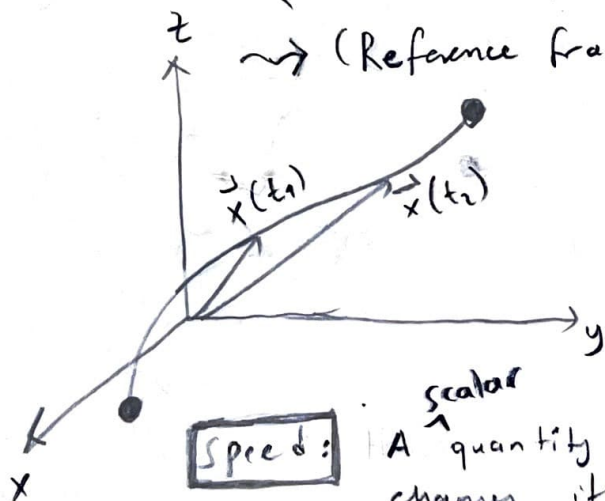
[2D]



e.g. snapshot of the "velocity" of the wind in a region

Lecture 1.2: (Ref. frame, position, velocity, speed, acceleration) and their derivatives and integrals

$\rightarrow$ (Reference frame)



$x(t)$: position of a particle w.r. to time

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$\rightarrow$ position vector

Speed: A ^{scalar} quantity representing how much a given particle changes its position in a given period of time.

e.g. $100 \frac{\text{km}}{\text{h}}$, $340 \frac{\text{m}}{\text{s}}$, $3 \times 10^8 \frac{\text{m}}{\text{s}}$

SI units: m/s

Dimensional $[L]/[T]$

$$S_{\text{avg}} = \frac{\Delta x}{\Delta t} \rightarrow \text{average speed, e.g., metro, ...}$$

$$S = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} \rightarrow \text{instantaneous speed: speed at a given time.}$$